Practice Exam - 1

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A 63-year old with hypertension, insulin-dependent diabetes mellitus, and hyperlipidemia is admitted for worsening heart failure. The patient has been admitted three times in the last six months. Home medications include aspirin 81 mg daily, atorvastatin 80 mg daily, long-acting insulin 30 units nightly, furosemide 20 mg daily, losartan 50 mg daily.

On examination, blood pressure is 150/80 mm Hg with a heart rate of 80 beats per minute. Body mass index is 38 kg/m2. Jugular venous pressure is 15 mm H2O. There are decreased breath sounds at the bases bilaterally. Abdomen is distended with a positive fluid wave. Cardiac examination is notable for regular rate and rhythm. There is a III/VI holosystolic murmur at the left lower sternal border that increases with inspiration. Lower extremities have 2+ pitting edema bilaterally.

Echocardiography demonstrates a left ventricular ejection fraction of 65%.

Which of the following is most likely to decrease rehospitalization for heart failure:

A. Therapy adjustment guided by pulmonary artery pressure monitoring

B. Transition of furosemide to torsemide

C. Addition of metoprolol succinate

D. Medication changes guided by daily weights

E. Addition of spironolactone

Commentary References

Testing Point: The Board-certified AHFTC specialist should know that implantable pulmonary artery pressure sensor monitoring has been demonstrated to reduce heart failure hospitalizations in patients with NYHA Class II or III regardless of ejection fraction.

Answer: A. Therapy adjustment guided by pulmonary artery pressure monitoring.

In the CHAMPION trial (the CardioMEMS Heart Sensor Allows Monitoring of Pressor to Improve Outcomes in NYHA Class III Heart Failure Patients), implantable pulmonary artery pressure sensor monitoring was shown to reduce the rate of heart failure hospitalizations. The Post-Approval Study also showed that this effect was consistent across ejection fraction groups.

Answer D: While decongestion is key for symptom relief and the patient may benefit from increased diuretic dose, it has not been demonstrated to decrease hospitalization, and weight has been shown to be a poor surrogate for filling pressures.

Answer B: Torsemide has theoretical benefits over furosemide but this was not borne out in the TRANSFORM-HF (Torsemide Comparison With Furosemide for Management of Heart Failure) Trial, which showed no statistical difference in outcomes for patients discharged from a heart failure hospitalization on furosemide compared to those discharged on torsemide. Patients with preserved ejection fraction were enrolled in TRANSFORM-HF.

Answer C: Beta-blockers have not been shown to reduce heart failure hospitalizations or survival in patients with preserved ejection fraction. Carvedilol may be reasonable to improve blood pressure control as this patient’s blood pressure is elevated. Spironolactone may be considered to decrease hospitalizations.

Answer E: Spironolactone may be considered in selected patients with HFpEF to reduce heart failure hospitalizations, particularly in those with elevated natriuretic peptides or recent hospitalization. However, the benefit is modest and less consistent across populations than the reduction in hospitalizations demonstrated with pulmonary artery pressure-guided management.